

# AI OPERATIONS COPILOT — PROJECT ROADMAP

*Detailed implementation checklist | Next.js + TypeScript + n8n + OpenAI + Supabase*

## PROJECT STRUCTURE

- AI Operations Copilot
- Admissions — core / fully functional
- Marketing — functional AI module
- Academic — functional AI module
- Operations — functional AI module
- Automations — workflow monitoring
- Training — employee enablement materials
- AI Tool Lab — tool experimentation
- AI Tool Evaluation — structured comparison and recommendation

## PHASE 0 — PRODUCT SPECIFICATION

1. Define the project objective: apply AI + automation to reduce repetitive work and improve employee productivity.
2. Set target company: education / training organization.
3. Define departments: Marketing, Admissions, Academic, Operations.
4. Define roles: Admin, Marketing Staff, Admissions Counselor, Teacher, Operations Manager.
5. List the main operational problems for each department.
6. Define AI use cases for each problem.
7. Mark each feature as Core / Secondary / Documentation-only.
8. Define MVP scope.
9. Create PRODUCT\_SPEC.md.
10. Create FEATURES.md.

Deliverable: clear product scope, feature list, roles, and MVP.

## PHASE 1 — SYSTEM ARCHITECTURE

11. Design the overall architecture: Frontend → Backend/API → n8n → OpenAI → Supabase → Gmail.
12. Define what Next.js handles.
13. Define what n8n handles.
14. Define what OpenAI handles.
15. Define what Supabase/PostgreSQL stores.
16. Define Gmail/email actions.
17. Document the complete Admissions data flow.
18. Design error/fallback paths.
19. Create an architecture diagram using Mermaid or Whimsical.
20. Create ARCHITECTURE.md.

Deliverable: architecture diagram + architecture documentation.

## PHASE 2 — DATABASE / CRM

Create these Supabase tables:

- leads: id, name, email, phone, source, course\_interest, budget, timeline, message, status, ai\_score, ai\_category, ai\_intent, ai\_summary, recommended\_action, created\_at.
- interactions: id, lead\_id, type, content, created\_at.
- follow\_up\_tasks: id, lead\_id, assigned\_to, priority, status, due\_at.
- automation\_logs: id, workflow\_name, lead\_id, event, status, error, created\_at.

21. Define primary keys and foreign keys.
22. Create indexes where useful.
23. Add RLS/security rules appropriate for the prototype.
24. Create seed/fake leads.
25. Implement CRUD/API access from Next.js.
26. Connect n8n to Supabase.
27. Test insert, update, query, and delete operations.

Deliverable: working database + CRM data structure.

## PHASE 3 — CORE ADMISSIONS AUTOMATION

Lead form fields:

- Name
- Email
- Phone
- Course
- Budget
- Timeline
- Message
- Source

Build workflow:

Lead Form → n8n Webhook → Validate → AI Analysis → Lead Scoring → CRM → HOT/WARM/COLD

28. Build the Next.js lead form.
29. Submit form data to an n8n webhook.
30. Validate required fields and email/phone formats.
31. Create the OpenAI prompt for structured lead analysis.
32. Force structured JSON output.
33. Generate score, category, intent, summary, and recommended action.
34. Define scoring rules and HOT/WARM/COLD thresholds.
35. Save the result to Supabase.
36. Create an automation log for every execution.
37. Create fake leads for HOT, WARM, COLD, invalid, and edge cases.
38. Test the complete flow end-to-end.

Example AI output:

```
{ "score": 92, "category": "HOT", "intent": "HIGH", "course": "IELTS", "timeline": "3 months", "summary": "...", "recommended_action": "Contact within 30 minutes" }
```

Deliverable: end-to-end AI lead classification workflow.

## PHASE 4 — AI RESPONSE & AUTOMATION ACTIONS

39. Generate AI lead summary.
40. Generate recommended counselor action.
41. Generate personalized lead response.
42. Update CRM.
43. Create follow-up task.
44. Send automated email.
45. Notify counselor through dashboard/email; Telegram or Slack is optional.
46. Log every workflow step.
47. Add error handling.
48. Add retry handling where appropriate.
49. Handle unexpected/invalid AI output.
50. Test success cases.
51. Test failure cases: AI failure, database failure, email failure, duplicate lead.

Core flow: Lead → AI → Score → CRM → AI Response → Email → Follow-up → Counselor Notification → Automation Log.

Deliverable: complete automated Admissions workflow.

## PHASE 5 — OPERATIONS DASHBOARD

Sidebar:

- Overview
- Admissions
- Marketing
- Academic
- Operations
- Automations
- Training

Overview metrics:

- Total leads
- HOT/WARM/COLD counts
- Automation runs
- Pending tasks
- Average response time
- Conversion rate

52. Build professional dashboard UI.
53. Connect dashboard to real Supabase data.
54. Build Admissions lead list.
55. Build lead detail page.
56. Show AI score, classification, summary, intent, and recommended action.
57. Show interaction history.
58. Show follow-up tasks.
59. Build Automation Logs page.
60. Show workflow, execution time, status, lead, and error.
61. Add filtering and search.
62. Add loading, empty, and error states.
63. Make the interface responsive and polished.

Deliverable: professional internal operations dashboard.

## PHASE 6 — DEPARTMENT AI MODULES

### 6.1 MARKETING

64. AI Content Generator: campaign + audience + platform + tone + objective → headlines, ad copy, CTA, variations.
65. Campaign Analyzer: campaign data → performance summary, best segment, weak segment, trends, recommendations.
66. Build UI forms.
67. Create and test prompts.
68. Use structured output where useful.
69. Create mock campaign data.
70. Validate inputs.
71. Display and allow copy/export of results.

### 6.2 ACADEMIC

72. AI Lesson Planner: grade + subject + topic + duration + objectives → lesson structure, activities, materials, homework.
73. AI Quiz Generator: topic + difficulty + count + question type → questions, answers, explanations.
74. Build UI.
75. Create prompts and structured outputs.
76. Test multiple subjects/difficulties.
77. Add copy/export functionality.
78. Include fact-checking / human-review guidance.

### 6.3 OPERATIONS

79. AI Report Generator: operational data → executive summary, key metrics, problems, trends, recommendations.
80. Create mock operational data.
81. Build input UI.
82. Connect OpenAI.
83. Display generated report.
84. Add copy/export.
85. Test different datasets.

Deliverable: AI modules demonstrating practical use across Marketing, Academic, and Operations.

## PHASE 7 — AI TOOL LAB + AI TOOL EVALUATION + EMPLOYEE ENABLEMENT

### 7.1 AI TOOL LAB

Goal: demonstrate hands-on research and experimentation with AI tools.

86. Select 5–8 relevant AI tools.
87. Group tools by use case: LLM, image, transcription, OCR, automation, research.
88. Define one practical business test for each tool.
89. Run the tests.
90. Record output quality.
91. Record speed.
92. Record cost/pricing.
93. Record ease of use.

94. Record integration capability.
95. Record limitations.
96. Capture screenshots/results.
97. Document findings.
98. Create a simple Tool Lab page if useful.

Deliverable: AI tool experimentation evidence + AI\_TOOL\_LAB.md.

## 7.2 AI TOOL EVALUATION

Goal: evaluate tools for actual business adoption, not just try them.

Score each tool from 1–5 on:

- Output Quality
- Accuracy
- Cost
- Speed
- Ease of Use
- Integration
- Security / Privacy
- Scalability

99. Define the business use case.
100. Define evaluation criteria and scoring method.
101. Run the test consistently.
102. Record scores and evidence.
103. Calculate total/weighted score if appropriate.
104. List strengths.
105. List weaknesses.
106. Decide: Recommended / Conditional / Not Recommended.
107. Explain the recommendation.
108. Document limitations and assumptions.

Deliverable: AI\_TOOL\_EVALUATION.md + comparison table + recommendations.

## 7.3 EMPLOYEE AI TRAINING

Do not claim real corporate training unless actually conducted. For the portfolio, design the training program/materials.

Marketing training:

- AI content creation
- Prompting
- Campaign analysis
- Output verification

Admissions training:

- AI lead scoring
- AI response usage
- Human review
- Escalation

Academic training:

- Lesson planning
- Quiz generation
- Fact checking
- Responsible AI

Operations training:

- AI reports
- Data analysis
- Workflow automation

109. Define learning objectives.
110. Create agenda.
111. Create examples.
112. Create exercises.
113. Create prompt templates.
114. Create practical assignments.
115. Create reference materials.
116. Create AI usage guidelines.
117. Define human-review requirements.

Deliverable: department-specific training materials.

## 7.4 INTERNAL AI WORKSHOP

Example 90-minute workshop: AI for Everyday Work

118. 0–10 min: Introduction and goals.
119. 10–25 min: Prompting basics.
120. 25–45 min: Department use cases.
121. 45–65 min: Live automation demo.
122. 65–80 min: Hands-on exercise.
123. 80–90 min: Q&A and best practices.
124. Create slides.
125. Create exercises.
126. Create facilitator/instructor guide.
127. Create feedback form.

Deliverable: workshop deck + exercises + instructor guide.

## 7.5 INTERNAL AI DOCUMENTATION / SOP

- AI Usage Guidelines.
  - AI Prompt Guide.
  - AI Lead Handling SOP.
  - AI Response Review SOP.
  - AI Automation User Guide.
  - AI Troubleshooting Guide.
  - AI Data Privacy Guidelines.
128. Define procedure for each SOP.
  129. Add screenshots.
  130. Add step-by-step instructions.
  131. Document exceptions.
  132. Define human escalation.
  133. Add troubleshooting steps.

134. Add examples.
135. Review documentation for clarity.

Deliverable: internal AI knowledge base / SOP collection.

## PHASE 8 — TESTING, POLISH & PORTFOLIO

### 8.1 TECHNICAL TESTING

136. Test valid inputs.
137. Test invalid inputs.
138. Test missing fields.
139. Test duplicate leads.
140. Test unexpected AI output.
141. Test OpenAI/API failure.
142. Test Supabase/database failure.
143. Test email failure.
144. Test n8n workflow failure.
145. Test retry behavior.

### 8.2 UI POLISH

146. Responsive layout.
147. Loading states.
148. Empty states.
149. Error states.
150. Form validation.
151. Consistent typography.
152. Consistent spacing.
153. Navigation quality.
154. Accessibility basics.
155. Professional internal-business appearance.

### 8.3 DOCUMENTATION

Create:

- README.md
- PRODUCT\_SPEC.md
- ARCHITECTURE.md
- AI\_DESIGN.md
- WORKFLOW.md
- AI\_TOOL\_LAB.md
- AI\_TOOL\_EVALUATION.md
- TRAINING.md

README sections:

- Overview
- Business Problem
- Solution
- Features
- Architecture
- Automation Workflow

- AI Implementation
- Screenshots
- Demo Video
- Tech Stack
- Setup
- Testing
- Limitations
- Future Improvements

Deliverable: tested, polished, documented portfolio project.

## PHASE 9 — DEMO & CV PACKAGE

### 9.1 SCREENSHOTS

- Overview dashboard
- Admissions
- AI lead analysis
- Automation workflow
- Marketing
- Academic
- Operations
- Training

### 9.2 DEMO VIDEO

156. Target length: 2–3 minutes.
157. Show dashboard.
158. Submit a fake lead.
159. Show AI analysis.
160. Show score/classification.
161. Show CRM update.
162. Show AI response.
163. Show email.
164. Show follow-up task.
165. Show automation log.
166. Briefly show Marketing, Academic, Operations, and Training.
167. Use OBS Studio.

### 9.3 GITHUB

- Source code.
- n8n workflows.
- Database schema.
- Documentation.
- Screenshots.
- Demo link.
- Setup instructions.
- .env.example.

### 9.4 CV

168. Use the project name.
169. List the tech stack.

170. Write 3 strong impact-focused bullets.
171. Add GitHub / portfolio / demo link.
172. Do not put the full architecture, documentation, or screenshots in the CV.

## RECOMMENDED TECH STACK

- Next.js + TypeScript
- Tailwind CSS + shadcn/ui
- n8n
- OpenAI API
- Supabase / PostgreSQL
- Gmail API
- GitHub
- Mermaid / Whimsical
- OBS Studio

## PRIORITY IF TIME IS LIMITED

- P0 — Phases 0–5: must complete.
- P1 — Phase 6: department AI modules.
- P2 — Phase 7: Tool Lab, Tool Evaluation, Training, Workshop, SOP.
- P3 — Phases 8–9: final polish, documentation, demo, CV package.
- Cut first: Telegram/Slack integration → Tool Lab UI → Training Center UI.
- Do NOT cut: n8n automation, AI classification, CRM, automated response, dashboard, automation logs.

## STRONGEST INTERVIEW DEMO

One complete Admissions flow:

Fake Lead → n8n → AI Analysis → Score → HOT/WARM/COLD → Supabase CRM → AI Response → Email → Follow-up → Counselor Notification → Automation Log → Dashboard

## IMPORTANT CV / INTERVIEW RULE

Present the project as a prototype/simulated internal AI operations platform. Do not claim real corporate AI training, workshops, or production deployment unless you actually performed them. For training/workshop work, use wording such as: “Designed an internal AI training program and workshop materials.”